

Computation of Greeks under rough Volterra stochastic volatility models using the Malliavin calculus approach

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Background

Background

For a discounted payoff

$$V^\theta = \mathbb{E} [e^{-rT} f(S_T^\theta)] ,$$

a Greek is the partial derivative:

$$\partial_\theta V^\theta = \partial_\theta \mathbb{E} [e^{-rT} f(S_T^\theta)]$$

The project compares three estimator families:

- Finite difference: generic, but sensitive to bump size.
- Pathwise derivative: efficient, but needs enough payoff smoothness.
- Malliavin / likelihood-ratio style: moves the derivative from the payoff to a random weight.

Finite Difference

$\partial_\theta V^\theta$ can be approximated by

$$\partial_\theta V^\theta \approx \frac{V^{\theta+h} - V^{\theta-h}}{2h}$$

This discretization has bias $O(h^2)$ and variance $O(h^{-2})$, and therefore - too large h creates discretization bias; - too small h amplifies Monte Carlo noise.

The second order derivative is approximated by

$$\partial_{\theta\theta}^2 V^\theta \approx \frac{V^{\theta+h} - 2V^\theta + V^{\theta-h}}{h^2}$$

This discretization has bias $O(h^2)$ and variance $O(h^{-4})$ and can be very unstable.

Pathwise Derivative

If the payoff function f is *nice* enough, we can swap the expectation and derivative

$$\partial_{\theta} \mathbb{E}[e^{-rT} f(S_T^{\theta})] = \mathbb{E}[\partial_{\theta} e^{-rT} f(S_T^{\theta})]$$

f is not nice if f has discontinuity.

The pathwise method is very efficient and accurate (0 bias), when applicable.

Likelihood Ratio Method and the Malliavin Approach

If the terminal state has density $p_\theta(x)$, then we have

$$\partial_\theta \mathbb{E}[f(X^\theta)] = \mathbb{E}[f(X^\theta) \frac{\partial_\theta p_\theta(X^\theta)}{p_\theta(X^\theta)}] = \mathbb{E}_\theta[f(X^\theta) \partial_\theta \log p_\theta(X^\theta)]$$

This is essentially computing a weighted $(\log p_\theta(X^\theta))$ mean of all the simulated payoffs.

It is also very efficient and accurate (0 bias), provided we know $p_\theta(x)$ explicitly.

Malliavin's method extends this method via the integration-by-parts formula:

$$\partial_\theta \mathbb{E}[f(X^\theta)] = \mathbb{E} \left[f(X^\theta) \cdot \delta \left(\frac{\partial_\theta X^\theta h}{\int_0^T h_t \mathcal{D}_t X^\theta dt} \right) \right]$$

Numerical Experiment

Numerical Experiment

We test Malliavin estimators for Δ and Γ under two rough Volterra volatility models:

α -RFSV / rough Bergomi

Rough Stein–Stein

$$V_t = V_0 \exp \left\{ \xi Y_t - \frac{1}{2} \alpha \xi^2 r(t) \right\}, \quad \alpha = 1.$$

$$dV_t = \kappa(\theta - V_t) dt + \nu dY_t.$$

In both cases,

$$Y_t = \int_0^t K_H(t, s) dZ_s, \quad Z_s = \rho W_s + \sqrt{1 - \rho^2} \widetilde{W}_s.$$

Estimator Comparison

- **Malliavin:** estimate Greeks as discounted payoffs multiplied by model-specific Malliavin weights.
- **Finite difference:** central difference in S_0 using common random numbers.
- **Pathwise:** included for vanilla Δ , where payoff differentiability permits it.

The convergence plots report running Monte Carlo means for Δ and Γ with 99% confidence intervals.

Experiment Setup

Experiment Setup

- Simulate terminal prices under both rough volatility models using the same rough-driver structure.
- Estimate Δ and Γ for vanilla, digital, and gap payoffs.
- Compare Malliavin weights with finite differences; add pathwise Δ where the payoff is smooth enough.

Simulation

- Shared parameters: $S_0 = 100$, $r = 0.05$, $T = 1$, $H = 0.15$, $\rho = -0.756$, $V_0 = 0.354$, $\varepsilon = 10^{-6}$.
- α -RFSV parameters: $\alpha = 1$, $\xi = 0.212$.
- Rough Stein–Stein parameters: $\kappa = 2.0$, $\theta = 0.5$, $\nu = 0.3$.
- Monte Carlo design: 200,000 paths, 2,000 time steps, fixed seed, and log-spaced checkpoints from 1,000 paths onward.
- Finite difference bump: $h = 10^{-5}$ applied to S_0 with the same simulated volatility and Brownian paths.

Payoffs

Continuous payoff

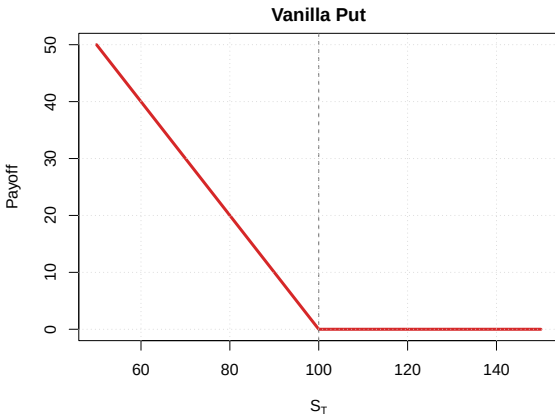
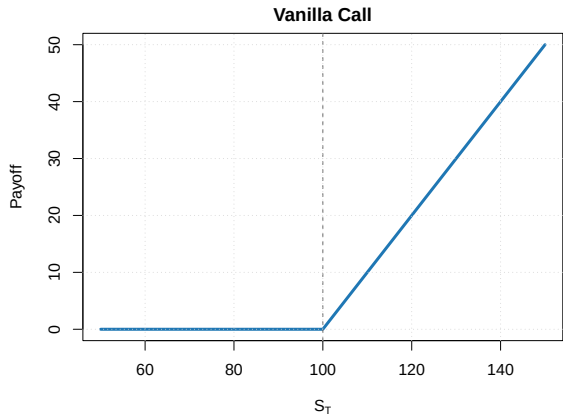
- Vanilla call/put with $K = 100$.
- Compare Malliavin, finite difference, and pathwise Δ .

Discontinuous payoff

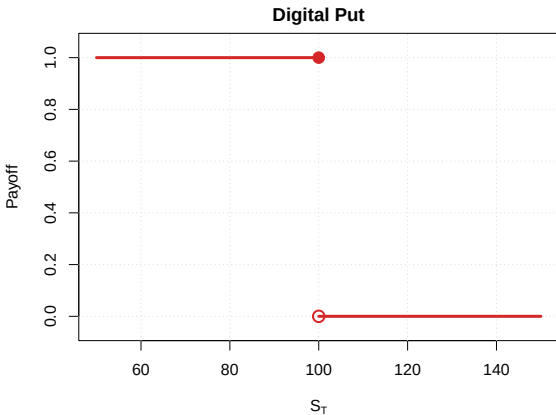
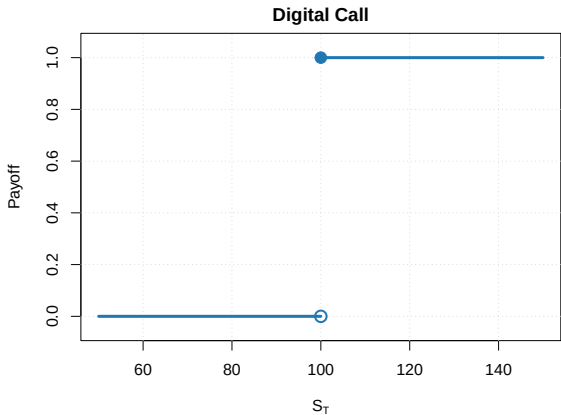
- Digital call/put with $K = 100$.
- Gap call with $(K_1, K_2) = (110, 100)$.
- Gap put with $(K_1, K_2) = (90, 100)$.
- Compare Malliavin against finite difference.

For rough Stein–Stein, paths with near-zero Malliavin denominator are filtered for numerical stability.

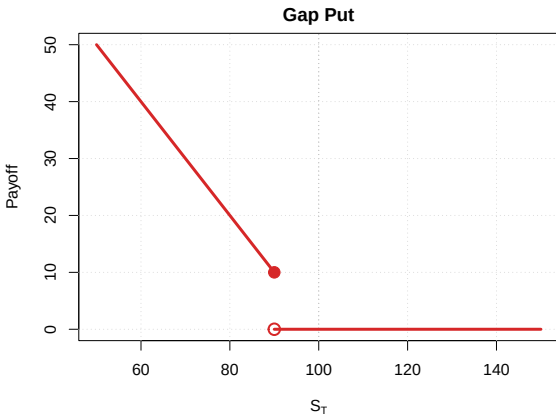
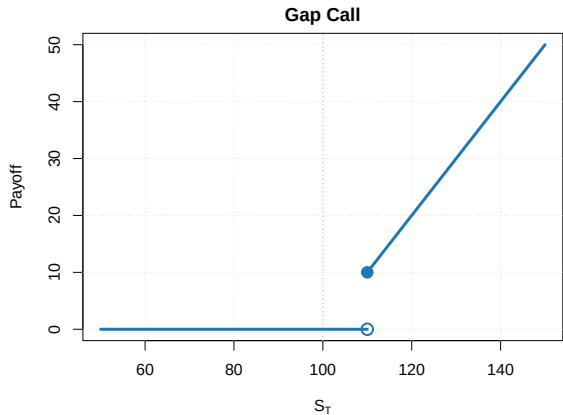
Vanilla Options



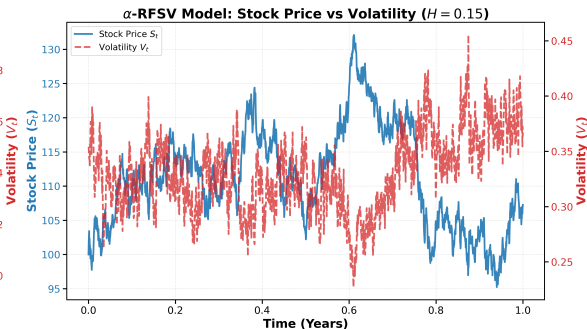
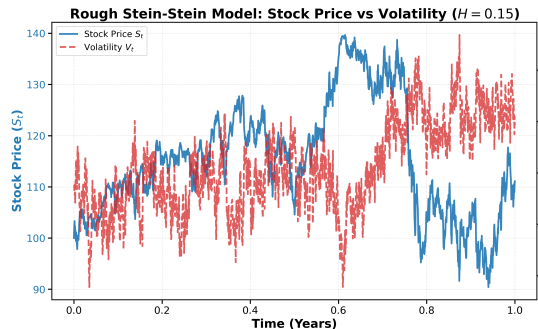
Digital Options



Gap Options



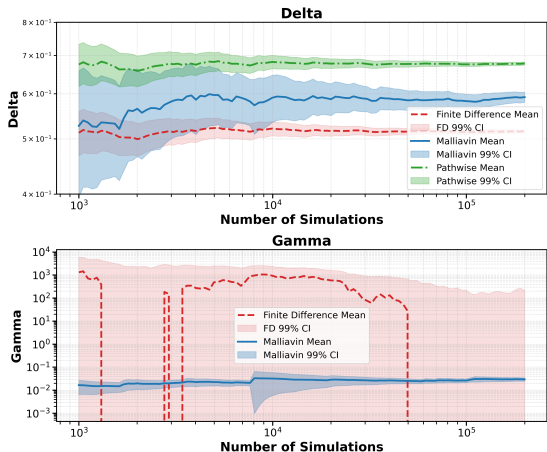
Simulated Price and Volatility Paths



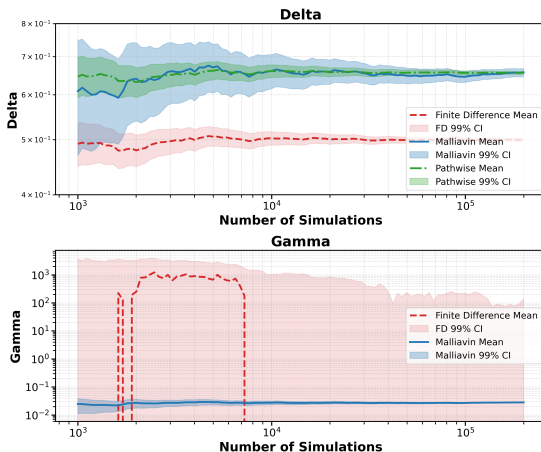
Results

Vanilla Call Options

Convergence Analysis: RSS European Vanilla Call (99% CI)

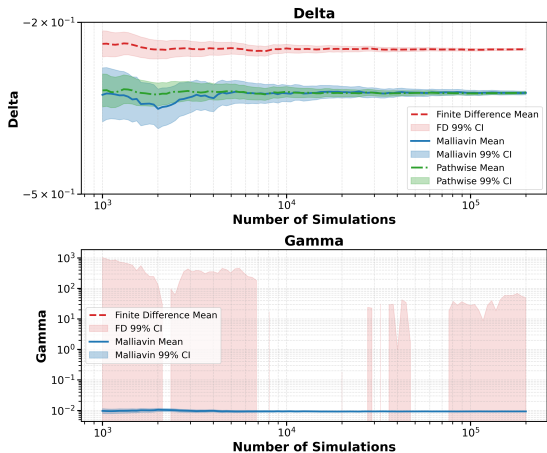


Convergence Analysis: α -RFSV European Vanilla Call (99% CI)

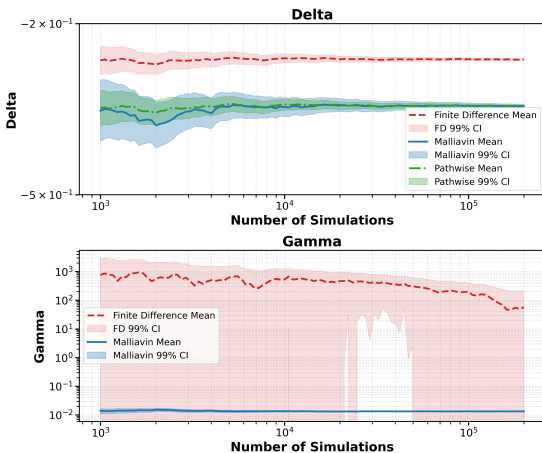


Vanilla Put Options

Convergence Analysis: RSS European Vanilla Put (99% CI)

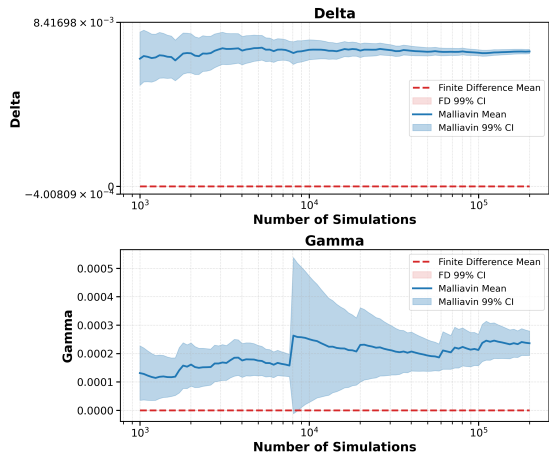


Convergence Analysis: α -RFSV European Vanilla Put (99% CI)

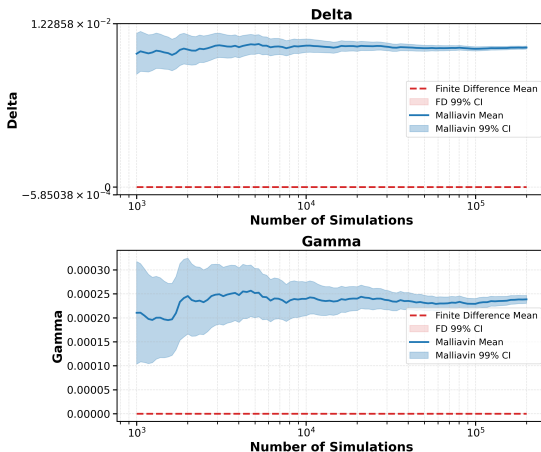


Digital Call Options

Convergence Analysis: RSS European Digital Call (99% CI)

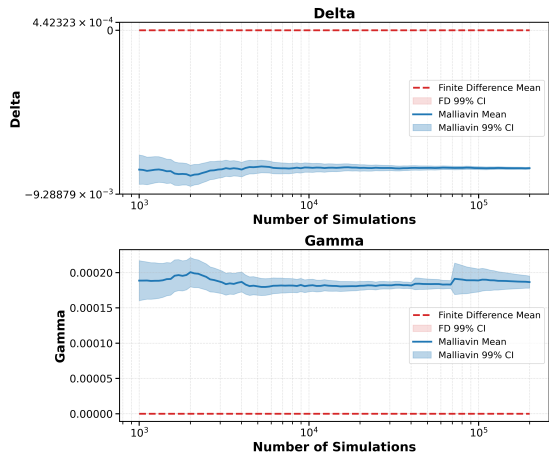


Convergence Analysis: α -RFSV European Digital Call (99% CI)

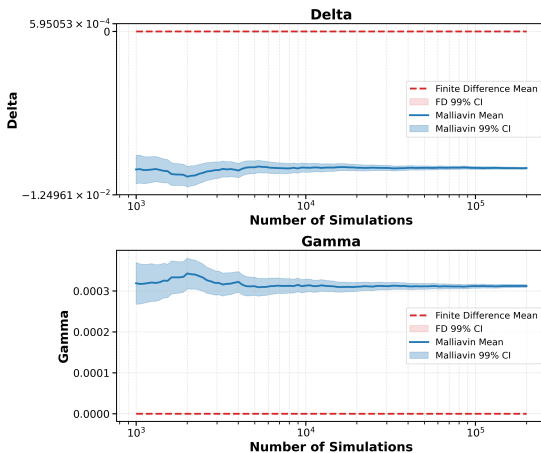


Digital Put Options

Convergence Analysis: RSS European Digital Put (99% CI)

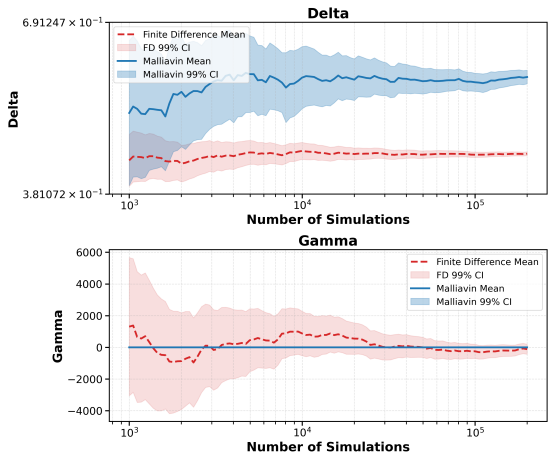


Convergence Analysis: α -RFSV European Digital Put (99% CI)

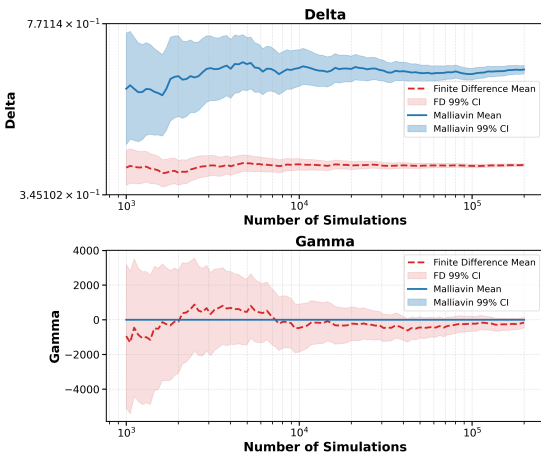


Gap Call Options

Convergence Analysis: RSS European Gap Call (99% CI)

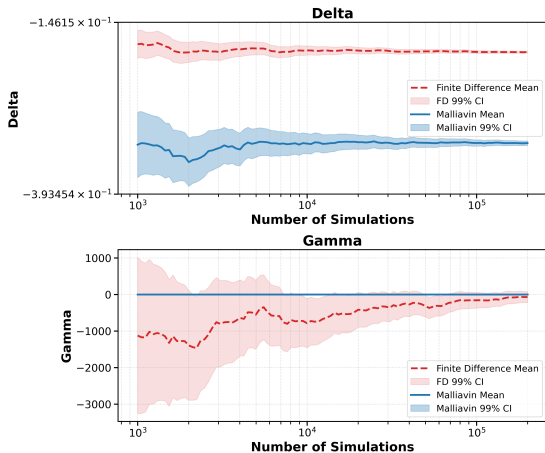


Convergence Analysis: α -RFSV European Gap Call (99% CI)

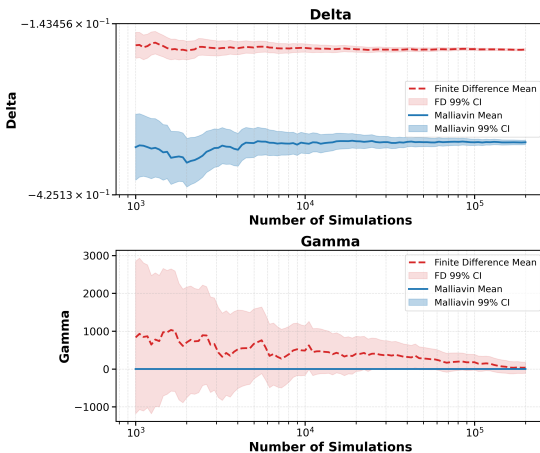


Gap Put Options

Convergence Analysis: RSS European Gap Put (99% CI)



Convergence Analysis: α -RFSV European Gap Put (99% CI)



Conclusion

Conclusion

- Malliavin provides unbiased estimates for second order Greeks where traditional methods fail.
- Finite difference becomes high variance and unstable for second order Greeks on exotic options with rough volatility.
- Finite difference also provides poor estimates for first order Greeks on exotic options with rough volatility.
- Pathwise derivative is efficient for smooth vanilla deltas, but fails for discontinuous payoffs.

References

- Al-Foraih, M., Bures, O., Pospisil, J., & Vives, J. (2025). *Computation of Greeks under rough Volterra stochastic volatility models using the Malliavin calculus approach*. arXiv:2312.00405v2.